

Course Code	
Course Name	C# Programming

Duration (in days)	13	Proficiency Level	Fundamental
Pre-requisites	C#	Target Audience	Campus Hires

Learning Outcome

At the end of the program, participants will be able to learn:

- .Net Core
- ADO.NET
- ADO.NET EF
- Dependency Injection in C#
- Unit Testing with NUnit/ Xunit
- Creating Mocks
- MoQ framework

Day-wise Session Plan

Day	Unit	Objective(s)	Hours
1	.Net Core Introduction	• .Net Core • .Net Core vs .Net Framework • Changes in .Net Core APIs • Building & Running .Net Core Applications using VS Code	4
		• Introduction to .NET CLI • .NET CLI Commands • The .Net Core Platform (Language Compilers, Core CLR and coreFx) • Building a basic .NET Core console application	4
2	Working with .Net Core	• Working with .Net Core Class Libraries • Using .Net Core Packages and Frameworks • Creating Web Applications in .Net Core • Running .Net Core Applications on Other platforms	8
3	ADO.NET Overview	• Introduction to Data Access Models • ADO.NET Architecture • ADO.NET Connected Architecture • ADO.NET Disconnected Architecture	1
	Getting Started with ADO.NET	• Setting Up Your Environment • Download Database • Create Project • Connect to A Data Source Using ADO.NET • SqlConnectionStringBuilder	1
	Retrieving Data from A Data Source	• Retrieving Data using Select Statement • DataReader	2
	CRUD Operations	• Insert • Update	4

		- Delete - Executing Stored Procedures	
4	CRUD Operations cont.	Stored Procedures with parameters	2
	Handling Errors in ADO.NET	- Catch Errors using try..catch block - The SqlException Class	2
	Disconnected Data Access	- DataSet,DataTable DataColumn, DataRow - SqlCommandBuilder - Update with SqlDataAdapter	4
5	Advanced Disconnected Data Access	- RowState, RowVersion,AcceptChanges,RejectChanges - Constraints Primary Key,Unique Key - Constraints –Foreign Key with Update and Delete rule,AcceptReject Rule - DataRelation - DataView - RowFilter and RowStateFilter - Sorting Views - Updating data through views - DataTableReader - Converting a Connected SqlDataReader to DataTable	4
	Managing Concurrency and Transactions	- Concurrency with WHERE clause - Concurrency with SqlDataAdapter - Concurrency with Timestamp - SqlTransaction - Commit or rollback a transaction, Savepoints - TransactionScope - Server Transactions	4
6	Entity Framework: Introduction	- Introduction - ORM Basics - Why Entity Framework - EF's Conceptual Mode - EF's Basic Workflow - Choosing DB First, Model First or Code First	4
	Database First Modeling	- Creating and inspecting the model and it's generated code - Model to Database Mappings - Updating Your Model when the Database Changes	4
7	Model First Modeling	- Why Model First - Creating a Model and your First Entity - Properties and Attributes of Entity Properties - Controlling Strings in Entities and Database Tables - Default Values, Spatial Data and Enums - Adding Relationships - Many to Many Relationships - Entity Container Properties - Generating the Database Schema when the Model or Database Changes	3
	Code First Modeling	- Why Code First? - The Domain Classes - Creating a DbContext Data Layer	5

		• Building, Debugging and Fixing the Code First Model with Configurations • Initializing a Database from the Model • Using Migrations to Update Database Schema when the Model Changes • Seeding a Database During a Migration • Using Data Annotations to Configure Mappings • Configuring Mappings with the Fluent API • Reverse Engineering a Database for Code First	
8	Interacting with Your Data Model	• The DbContext API • Your First Queries • Inserting, Updating and Deleting Data • Building and Inserting Graphs • Querying with LINQ Expressions or Methods • Loading Related Data • Working with Stored Procedures	8
9	Dependency Injection	• What are Dependencies • Definition of IoC and DI • Refactoring to a Better Design • Dependency Injection Techniques • Constructor Injection • Property Injection • Interface Injection • DI Frameworks for .Net • Architectural Implications • DI and IoC-Containers • Building a Simple IoC-Container • Service Locator	8
10	Unit Testing Introduction	• Getting Started • Your First Unit Test • Naming Conventions • Running and Debugging Unit Tests • Benefits of Unit Testing	4
	Testing Frameworks for C#	• NUnit • Xunit • MSTest • Nunit Vs Xunit vs MSTest • Creating & Running your first Nunit test	4
11	Asserts	• Assert Introduction • Arrange-Act-Assert • Running Tests from the Console • SetUp and TearDown • OneTimeSetUp and OneTimeTearDown • Parameterized Unit Tests • Code Coverage	8

12	Mocks	• Overview • Mocks, Stubs, Fakes, Moles — What Are They All For? • Static Fakes • Dynamic Fakes • Stubs • Mocks	8
13	MoQ Framework	• Get Started with Moq • Mocking Methods • Argument-Dependent Mocking • Exceptions and Return Values • Mocking Properties • Value Tracking • Mocking Events • Callbacks • Verification • Behaviour Customization	6
	Unit testing best practices with .NET Core and .NET Standard.	• Why unit test? • Characteristics of a good unit test • Code coverage	2